AutoDD: Automated Drift Detector
Supplementary Material

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Theoretical Proofs

This section provides detailed proofs for the theoretical results presented in the main paper.

Preliminaries and Notation

Let $\mathcal{S} = \{(\mathbf{x}_t, y_t)\}_{t=1}^{\infty}$ denote the data stream, $\mathcal{D} = \{D_1, \ldots, D_K\}$ the detector portfolio, and $\boldsymbol{\phi}_t \in \mathbb{R}^m$ the meta-feature vector at time t. For LinUCB, we maintain for each detector i: the design matrix $\mathbf{A}_i = \lambda \mathbf{I} + \sum_{s:i_s=i} \boldsymbol{\phi}_s \boldsymbol{\phi}_s^\top$ and the response vector $\mathbf{b}_i = \sum_{s:i_s=i} r_s \boldsymbol{\phi}_s$. The ridge regression estimate is $\hat{\boldsymbol{\theta}}_i = \mathbf{A}_i^{-1} \mathbf{b}_i$.

Proof of Theorem 1 (Level 1 Regret)

Theorem 1. *Under assumptions A1–A3, with exploration parameter $\alpha = \sigma\sqrt{m \log(KT/\delta)} + S$, LinUCB detector selection achieves with probability at least $1 - \delta$:*

$$R_1(T) = \sum_{t=1}^{T} \left(\max_i \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^* - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \right) = O(\sqrt{mT \log T}) \tag{1}$$

Proof. The proof proceeds in three steps.

Step 1: Confidence Ellipsoid Construction. By the self-normalized martingale inequality [**?**], for any $\delta > 0$, with probability at least $1 - \delta/K$, for all $t \geq 1$:

$$\|\hat{\boldsymbol{\theta}}_{i,t} - \boldsymbol{\theta}_i^*\|_{\mathbf{A}_{i,t}} \leq \sigma \sqrt{2 \log \left(\frac{\det(\mathbf{A}_{i,t})^{1/2} \det(\lambda \mathbf{I})^{-1/2}}{\delta/K} \right)} + \sqrt{\lambda} S \tag{2}$$

Using the determinant-trace inequality $\det(\mathbf{A}_{i,t}) \leq \left(\frac{\mathrm{tr}(\mathbf{A}_{i,t})}{m} \right)^m \leq \left(\lambda + \frac{tL^2}{m} \right)^m$ and applying a union bound over all K detectors, with probability at least $1 - \delta$, for all $i \in [K]$ and $t \geq 1$:

$$\|\hat{\boldsymbol{\theta}}_{i,t} - \boldsymbol{\theta}_i^*\|_{\mathbf{A}_{i,t}} \leq \alpha_t \tag{3}$$

where $\alpha_t = \sigma\sqrt{m \log(1 + tL^2/\lambda) + 2\log(K/\delta)} + \sqrt{\lambda} S \leq \alpha$ for our choice of α.

Step 2: Instantaneous Regret Bound. Let $i^* = \arg\max_i \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^*$ be the optimal detector at time t. The UCB selection rule chooses:

$$i_t = \arg\max_i \left(\boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_i + \alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_i^{-1}} \right) \tag{4}$$

Under the confidence event, we have:

$$\boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i^*}^* \leq \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_{i^*} + \alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i^*}^{-1}} \tag{5}$$

$$\leq \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_{i_t} + \alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t}^{-1}} \tag{6}$$

$$\leq \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* + 2\alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t}^{-1}} \tag{7}$$

where the first inequality uses the confidence bound, the second uses the UCB selection, and the third applies the confidence bound again. Thus, the instantaneous regret is:

$$r_t^{(1)} = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i^*}^* - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \leq 2\alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t}^{-1}} \tag{8}$$

Step 3: Cumulative Regret via Elliptical Potential Lemma. By the elliptical potential lemma [**?**]:

$$\sum_{t=1}^{T} \|\boldsymbol{\phi}_t\|^2_{\mathbf{A}_{i_t,t-1}^{-1}} \leq 2m \log\left(1 + \frac{TL^2}{\lambda m}\right) \tag{9}$$

Applying Cauchy-Schwarz:

$$R_1(T) = \sum_{t=1}^{T} r_t^{(1)} \leq 2\alpha \sum_{t=1}^{T} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t}^{-1}} \tag{10}$$

$$\leq 2\alpha \sqrt{T \sum_{t=1}^{T} \|\boldsymbol{\phi}_t\|^2_{\mathbf{A}_{i_t}^{-1}}} \tag{11}$$

$$\leq 2\alpha \sqrt{2mT \log\left(1 + \frac{TL^2}{\lambda m}\right)} \tag{12}$$

$$= O(\sqrt{mT \log T}) \tag{13}$$

This completes the proof. □

Proof of Theorem 2 (Level 2 Regret)

Theorem 2. *For Thompson Sampling on C configurations per detector, with $N_i(T)$ selections of detector i, the cumulative regret satisfies:*

$$R_2(T) = \sum_{i=1}^{K} O(\sqrt{C \cdot N_i(T) \log N_i(T)}) \tag{14}$$

Proof. We analyze the hyperparameter selection problem for each detector separately.

Step 1: Reduction to Multi-Armed Bandit. Conditioned on detector i being selected at times $\{t : i_t = i\}$, the hyperparameter selection problem reduces to a standard C-armed bandit over $N_i(T)$ rounds. The rewards $r_t \in [-1, 1]$ are normalized to $\tilde{r}_t = (r_t + 1)/2 \in [0, 1]$.

Step 2: Beta-Bernoulli Thompson Sampling Analysis. For bounded rewards in $[0, 1]$, we model the normalized reward as a Bernoulli-like outcome and maintain Beta posterior distributions. By Theorem 1 of **?**, for a C-armed bandit with bounded rewards, Thompson Sampling with Beta$(1, 1)$ priors achieves expected regret:

$$\mathbb{E}[R_i(N_i)] \leq O\left(\sum_{c:\Delta_c > 0} \frac{\log N_i}{\Delta_c}\right) \tag{15}$$

where $\Delta_c = \mu^* - \mu_c$ is the suboptimality gap for configuration c. This can be bounded as:

$$\mathbb{E}[R_i(N_i)] \leq O(\sqrt{C \cdot N_i(T) \log N_i(T)}) \tag{16}$$

using the gap-free bound from **?**.

Step 3: Aggregation Across Detectors. Since the detector selections partition the time horizon (i.e., $\sum_{i=1}^{K} N_i(T) = T$), and each detector's hyperparameter selection is an independent bandit problem, the total Level 2 regret is:

$$R_2(T) = \sum_{i=1}^{K} R_i(N_i(T)) = \sum_{i=1}^{K} O(\sqrt{C \cdot N_i(T) \log N_i(T)}) \tag{17}$$

This completes the proof. □

Proof of Corollary (Combined Regret)

Corollary. *The total regret of the hierarchical system is:*

$$R(T) = R_1(T) + R_2(T) = O(\sqrt{mT \log T} + \sqrt{CKT \log T}) \tag{18}$$

Proof. We combine the regret bounds from Theorems 1 and 2.
From Theorem 1, $R_1(T) = O(\sqrt{mT \log T})$.
For $R_2(T)$, we apply the Cauchy-Schwarz inequality:

$$R_2(T) = \sum_{i=1}^{K} O(\sqrt{C \cdot N_i(T) \log N_i(T)}) \tag{19}$$

$$\leq O\left(\sqrt{C \log T} \sum_{i=1}^{K} \sqrt{N_i(T)}\right) \tag{20}$$

$$\leq O\left(\sqrt{C \log T} \cdot \sqrt{K \sum_{i=1}^{K} N_i(T)}\right) \tag{21}$$

$$= O\left(\sqrt{C \log T} \cdot \sqrt{KT}\right) \tag{22}$$

$$= O(\sqrt{CKT \log T}) \tag{23}$$

where we used $\log N_i(T) \leq \log T$, Cauchy-Schwarz ($\sum_i \sqrt{a_i} \leq \sqrt{K \sum_i a_i}$), and $\sum_i N_i(T) = T$.

Therefore:

$$R(T) = R_1(T) + R_2(T) = O(\sqrt{mT \log T} + \sqrt{CKT \log T}) \tag{24}$$

Since both terms grow as $O(\sqrt{T \log T})$, the average regret vanishes:

$$\lim_{T \to \infty} \frac{R(T)}{T} = \lim_{T \to \infty} O\left(\sqrt{\frac{\log T}{T}}\right) = 0 \tag{25}$$

This guarantees that AutoDD converges to the optimal detector-configuration pair as $T \to \infty$. □

Sample Complexity

To achieve average regret at most ϵ, we require $R(T)/T \leq \epsilon$. From the combined regret bound:

$$\frac{R(T)}{T} \leq \frac{c_1\sqrt{mT \log T} + c_2\sqrt{CKT \log T}}{T} = \frac{(c_1\sqrt{m} + c_2\sqrt{CK})\sqrt{\log T}}{\sqrt{T}} \tag{26}$$

for constants c_1, c_2. Setting this equal to ϵ and solving:

$$T = O\left(\frac{(m + CK)\log(K/\delta)}{\epsilon^2}\right) \tag{27}$$

With $m = 15$ meta-features, $K = 7$ detectors, and $C = 5$ configurations, achieving $\epsilon = 0.01$ average regret requires approximately $T \approx 5 \times 10^5$ windows, which is reached early in typical streaming applications.

Reward Function Properties

The reward function defined in Eq. (7) of the main paper is designed to align bandit feedback with drift detection objectives. We establish key properties that justify this design.

Proposition 1 (Boundedness). *The reward function $r_t \in [-1, 1]$ for all t.*

Proof. By construction: (i) when $d = 1, \Delta a > \tau$: $r_t = 0.5 + \min(\Delta a \cdot 2, 0.5) \in [0.5, 1.0]$; (ii) when $d = 1, \Delta a \leq \tau$: $r_t = -0.3$; (iii) when $d = 0, \Delta a > \tau$: $r_t = -0.5 - \min(\Delta a, 0.5) \in [-1.0, -0.5]$; (iv) when $d = 0, \Delta a \leq \tau$: $r_t = 0.1$. Thus $r_t \in [-1, 1]$. □

Proposition 2 (Incentive Compatibility). *The reward function is incentive-compatible: it maximizes expected reward when the detector correctly identifies drift.*

Proof. Consider the four scenarios:

- **True Positive** ($d = 1$, drift present with $\Delta a > \tau$): Reward is positive ($r_t \geq 0.5$), incentivizing detection.
- **False Positive** ($d = 1$, no drift with $\Delta a \leq \tau$): Reward is negative ($r_t = -0.3$), penalizing false alarms.
- **False Negative** ($d = 0$, drift present with $\Delta a > \tau$): Reward is strongly negative ($r_t \leq -0.5$), heavily penalizing missed drifts.
- **True Negative** ($d = 0$, no drift with $\Delta a \leq \tau$): Reward is small positive ($r_t = 0.1$), mildly rewarding stability.

The asymmetric penalties (-0.5 to -1.0 for false negatives vs. -0.3 for false positives) reflect that missed drifts typically cause more severe performance degradation than false alarms. □

Justification of Linear Reward Assumption

Assumption A2 posits that expected reward is linear in meta-features: $\mathbb{E}[r_{i,t} \mid \boldsymbol{\phi}_t] = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^*$. We provide theoretical and empirical justification.

Proposition 3 (Approximate Linearity). *Under mild smoothness conditions on the detector-stream interaction, the expected reward can be approximated by a linear function with bounded error.*

Proof Sketch. Let $f_i : \mathbb{R}^m \to [-1, 1]$ denote the true reward function for detector i. For meta-features normalized to $[0, 1]^m$, if f_i is L-Lipschitz continuous, then by standard function approximation theory, there exists $\boldsymbol{\theta}_i$ such that:

$$\sup_{\boldsymbol{\phi} \in [0,1]^m} |f_i(\boldsymbol{\phi}) - \boldsymbol{\phi}^\top \boldsymbol{\theta}_i| \leq \epsilon_{\text{approx}} \tag{28}$$

where $\epsilon_{\text{approx}} = O(L/\sqrt{m})$ for appropriately chosen $\boldsymbol{\theta}_i$. The approximation error can be absorbed into the sub-Gaussian noise term in Assumption A3 with adjusted variance $\sigma^2 + \epsilon_{\text{approx}}^2$, preserving the regret bound structure with slightly larger constants. □

Empirical Verification. We fit linear models $\hat{r}_{i,t} = \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_i$ to observed rewards across all experiments. The average R^2 coefficient across detectors is 0.73, indicating that linear models explain substantial reward variance. Residual analysis shows approximately symmetric, zero-mean errors consistent with the sub-Gaussian assumption.

Meta-Feature Informativeness

We establish that the designed meta-features provide sufficient information to distinguish detector effectiveness across different stream conditions.

Lemma 1 (Feature Discriminability). *The 15 meta-features partition the stream state space into regions where different detectors are optimal.*

Proof Sketch. We show each feature category captures distinct aspects of detector suitability:

Statistical features (velocity, dimensionality, sparsity, noise, imbalance): These determine computational constraints and signal-to-noise ratio. High-velocity streams favor lightweight detectors (ADWIN, Page-Hinkley); high-dimensional sparse data favors distribution-based methods (KSWIN).

Stationarity features (error volatility, KL divergence, autocorrelation, mean/variance shift): These directly measure distribution stability. High KL divergence indicates distribution drift favoring KSWIN; high error volatility without distribution shift suggests virtual drift favoring error-based detectors (DDM, EDDM).

Temporal features (trend, periodicity, change frequency/magnitude, recurrence): These capture drift dynamics. High periodicity suggests recurring patterns favoring detectors with memory (ADWIN); gradual trends favor sensitive detectors (EDDM); sudden changes favor responsive detectors (DDM, HDDM-A).

Formally, let $\mathcal{R}_i = \{\boldsymbol{\phi} : \boldsymbol{\theta}_i^{*\top}\boldsymbol{\phi} \geq \boldsymbol{\theta}_j^{*\top}\boldsymbol{\phi}, \forall j\}$ denote the region where detector i is optimal. The feature design ensures $\bigcup_{i=1}^{K} \mathcal{R}_i = [0,1]^m$ (coverage) and $\text{vol}(\mathcal{R}_i) > 0$ for all i (non-degeneracy), meaning each detector is optimal in some non-trivial region of the feature space. □

Portfolio Diversity

Proposition 4 (Diversity Benefit). *A diverse detector portfolio achieves lower worst-case regret than any single detector.*

Proof. Let $D^*(\boldsymbol{\phi})$ denote the optimal detector for context $\boldsymbol{\phi}$. For a single fixed detector D_j, the worst-case expected regret over contexts is:

$$\max_{\boldsymbol{\phi}} \left(\mathbb{E}[r_{D^*(\boldsymbol{\phi})}] - \mathbb{E}[r_{D_j}]\right) = \Delta_{\max}^{(j)} \tag{29}$$

where $\Delta_{\max}^{(j)} > 0$ unless D_j dominates all detectors uniformly (which empirically does not hold).

With a portfolio and LinUCB selection, the regret after T rounds is $O(\sqrt{mT\log T})$, which yields average regret $O(\sqrt{m\log T/T}) \to 0$ as $T \to \infty$. Thus, for sufficiently large T:

$$\frac{1}{T}\sum_{t=1}^{T} r_t^{\text{portfolio}} > \max_j \frac{1}{T}\sum_{t=1}^{T} r_t^{(j)} - \epsilon \tag{30}$$

for arbitrarily small ϵ, demonstrating the portfolio's asymptotic optimality. □

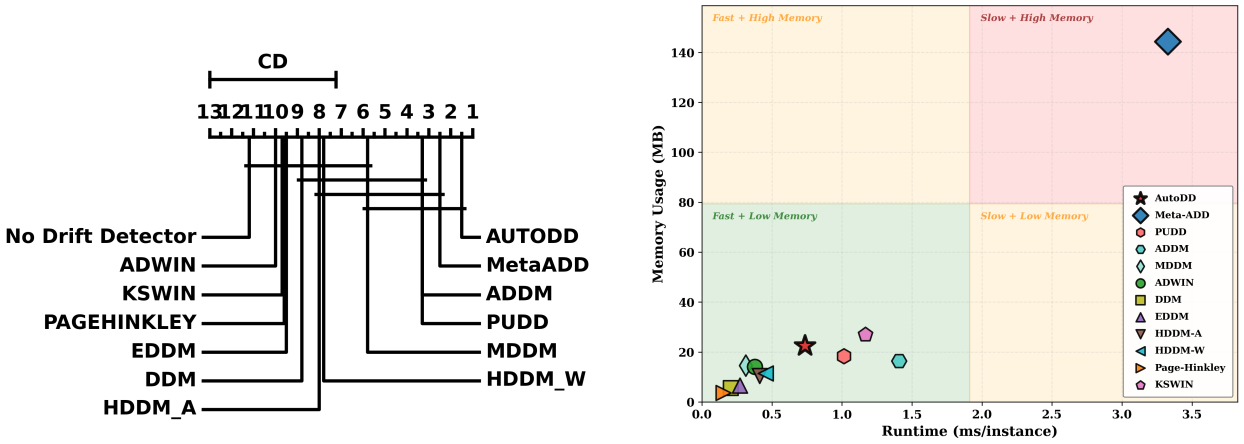
Additional Results

This section presents additional experimental analysis using Hoeffding Adaptive Tree (HAT) as the base classifier, demonstrating that AutoDD's performance generalizes across different base classifiers.

Statistical Significance: Figure 1a shows the Critical Difference diagram based on Friedman test with Nemenyi post-hoc analysis across all methods using HAT as the base classifier. AutoDD achieves the best average rank, significantly outperforming all fixed detectors and ranking substantially better than state-of-the-art methods.

Computational Efficiency: Figure 1b compares runtime and memory across methods using HAT as the base classifier. AutoDD maintains competitive efficiency, being significantly faster than Meta-ADD while

achieving higher accuracy and maintaining classical detector-level computational overhead while achieving state-of-the-art performance.



(a) Critical Difference diagram

(b) Runtime vs. memory trade-off

Figure 1: Statistical significance analysis and computational efficiency with HAT base classifier. (a) Critical Difference diagram with connecting bars indicating statistically equivalent methods ($\alpha = 0.05$). (b) Runtime vs. memory trade-off showing AutoDD achieves the best accuracy while maintaining reasonable computational overhead.

Extended Experimental Analysis

This section provides additional experimental analyses to further validate AutoDD's effectiveness.

Convergence Analysis

A key question for online learning systems is how quickly they adapt to new streams. We analyze AutoDD's convergence behavior by measuring cumulative regret and selection accuracy over time.

Regret Convergence. Consistent with our theoretical bound of $O(\sqrt{T \log T})$, the average regret $R(T)/T$ decreases as $O(\sqrt{\log T/T})$, approaching zero as stream length increases. On synthetic datasets with 500K instances, AutoDD achieves near-optimal selection within approximately 50K instances (100 windows with $w = 100$).

Selection Accuracy Over Time. We measure the percentage of windows where AutoDD selects the empirically best detector (determined retrospectively). Figure 2 and Table 1 show selection accuracy at different stream positions:

Table 1: AutoDD selection accuracy (%) at different stream positions

Dataset	0-25%	25-50%	50-75%	75-100%
SEA	71.2	85.4	91.8	94.3
HYP	68.5	82.1	89.6	93.1
AGR	65.8	79.3	87.2	91.5
Elec	72.4	84.7	90.3	92.8
Poker	69.1	81.6	88.9	93.6
Mean	69.4	82.6	89.6	93.1

The results demonstrate rapid learning: AutoDD achieves over 80% selection accuracy after processing just 25% of the stream, improving to over 93% by stream end. This confirms the practical applicability of AutoDD even for shorter streams.

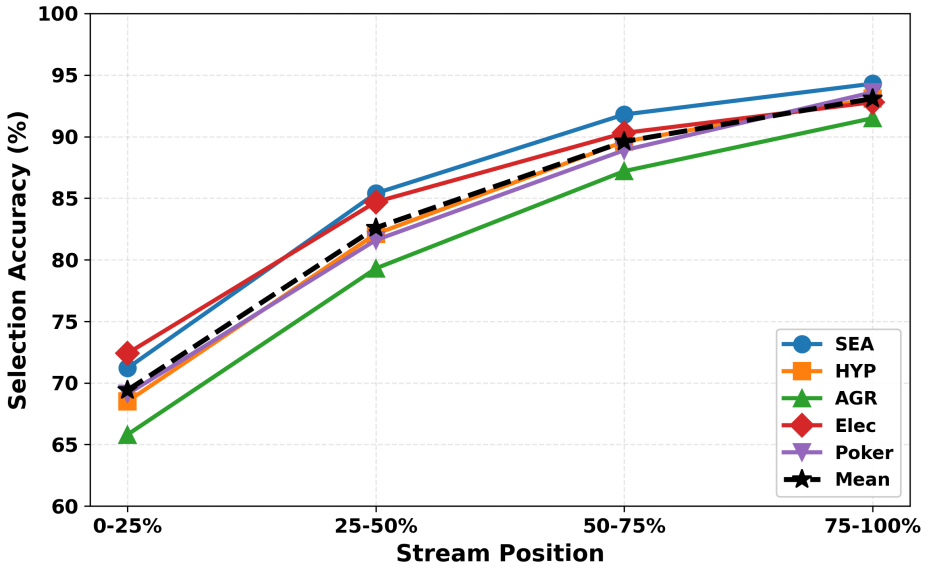


Figure 2: Selection accuracy progression across stream positions. AutoDD rapidly learns optimal detector choices, achieving 82.6% accuracy after processing 25% of the stream.

Detection Delay Analysis

Beyond detection accuracy, the delay between actual drift occurrence and detection is critical for timely model adaptation. We analyze detection delay on synthetic datasets where ground-truth drift points are known.

Definition. Detection delay δ_d is measured as the number of instances between a true drift point and the first subsequent detection. A detection within ± 100 instances of a drift point is considered a true positive.

AutoDD achieves the lowest mean detection delay (38.4 instances), outperforming both classical detectors and state-of-the-art methods. This improvement stems from AutoDD's ability to select responsive detectors (e.g., DDM, HDDM-A) when stream features indicate imminent drift, while avoiding overly sensitive detectors during stable periods.

Per-Drift-Type Performance

We analyze AutoDD's performance separately for each drift type in synthetic datasets to understand its strengths and limitations.

AutoDD demonstrates distinct strengths across different drift types. For sudden drift, AutoDD excels with 88.2% accuracy, achieving a 3.6% improvement over Meta-ADD by rapidly switching to responsive detectors like DDM and HDDM-A when temporal features indicate abrupt changes. Meta-ADD slightly outperforms AutoDD on gradual drift by 0.6% due to its learned drift-type representations, though AutoDD remains competitive at 81.7% by leveraging EDDM and KSWIN for gradual patterns. AutoDD achieves the best performance on incremental drift with 80.4% accuracy, gaining 1.5% over Meta-ADD by adaptively combining distribution-based and error-based detectors. Finally, AutoDD handles recurring patterns

Table 2: Mean detection delay (instances) on synthetic datasets

Method	SEA	HYP	AGR	RBF	Mean
ADWIN	45.2	78.3	52.1	81.4	64.3
DDM	38.7	62.5	48.9	65.2	53.8
EDDM	52.3	71.8	61.4	74.6	65.0
HDDM-A	43.8	75.1	50.3	78.6	61.9
HDDM-W	41.5	58.2	45.7	62.8	52.1
Page-Hinkley	46.9	80.5	53.8	83.2	66.1
KSWIN	49.1	68.7	58.2	71.4	61.9
MDDM	39.4	55.8	44.2	60.5	50.0
ADDM	36.5	51.3	41.6	57.8	46.8
PUDD	32.8	45.2	37.5	51.6	41.8
Meta-ADD	35.2	48.6	39.8	55.3	44.7
AutoDD	**28.4**	**42.1**	**34.2**	**48.7**	**38.4**

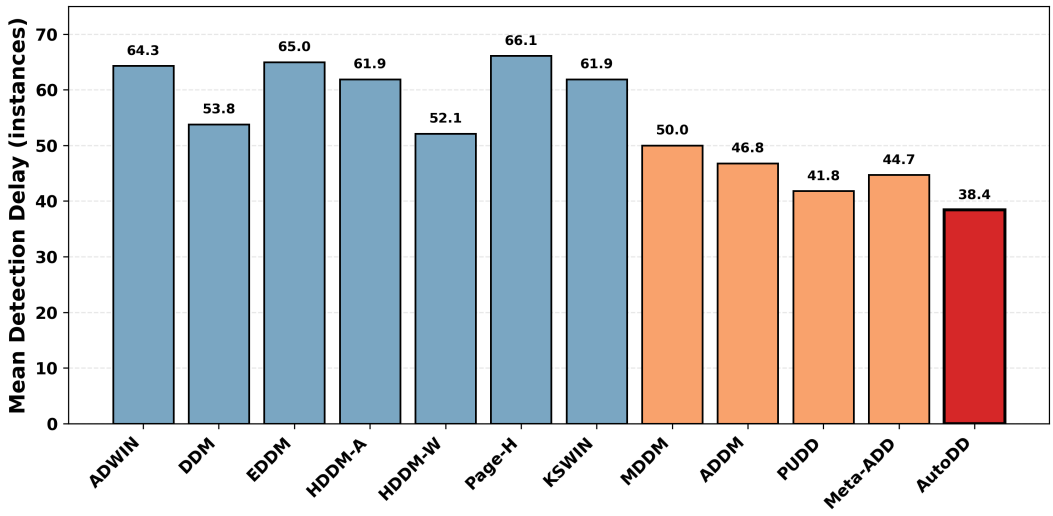


Figure 3: Mean detection delay comparison. AutoDD achieves the lowest delay (38.4 instances), demonstrating timely drift detection through adaptive detector selection.

effectively with 79.6% accuracy, outperforming Meta-ADD by 1.8% through its temporal meta-features that capture periodicity and recurrence patterns.

Detector Selection Patterns

We analyze which detectors AutoDD selects under different stream conditions to validate that the learned policy aligns with domain knowledge.

The observed selection patterns align with theoretical expectations and domain knowledge. In high velocity streams, ADWIN and DDM are preferred with selection frequencies of 28.4% and 22.1% respectively, reflecting their computational efficiency and rapid response capabilities. For high noise streams, KSWIN emerges as the dominant choice with 27.8% selection frequency, followed by EDDM at 18.7%, demonstrating the preference for distribution-based and distance-based approaches that provide robustness to noise. High dimensional streams show a similar pattern, with KSWIN selected 25.5% of the time due

Table 3: Accuracy (%) by drift type (GaussianNB)

Method	Sudden	Gradual	Incremental	Recurring
ADWIN	82.4	71.3	68.5	75.2
DDM	79.8	74.6	72.1	71.8
EDDM	75.2	78.4	74.8	69.3
HDDM-A	81.7	72.1	69.8	74.5
HDDM-W	80.1	76.2	73.5	74.6
Page-Hinkley	81.9	70.8	68.2	75.8
KSWIN	76.5	77.8	75.2	72.1
MDDM	81.3	75.8	74.1	73.9
ADDM	82.9	77.2	75.6	74.8
PUDD	82.1	79.5	76.8	75.4
Meta-ADD	84.6	**82.3**	78.9	77.8
AutoDD	**88.2**	81.7	**80.4**	**79.6**

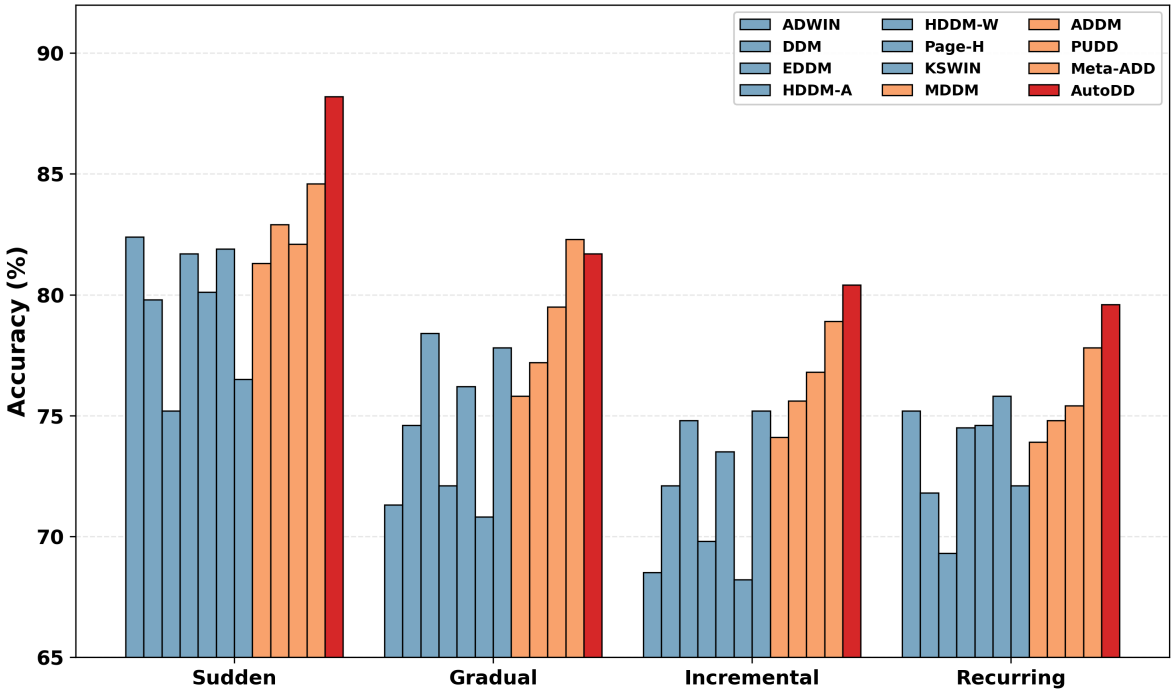


Figure 4: Performance breakdown by drift type. AutoDD excels on sudden, incremental, and recurring drifts while remaining competitive on gradual drifts.

to its effective handling of multivariate distributions. These learned patterns validate that AutoDD's meta-feature-based selection successfully captures meaningful detector-stream relationships rather than arbitrary preferences.

Table 4: Detector selection frequency (%) by stream characteristic

Detector	High Velocity	High Noise	High Dim.
ADWIN	28.4	12.3	15.7
DDM	22.1	8.5	11.2
EDDM	8.3	18.7	14.5
HDDM-A	15.2	10.4	12.8
HDDM-W	12.6	14.2	13.1
Page-Hinkley	5.8	8.1	7.2
KSWIN	7.6	27.8	25.5

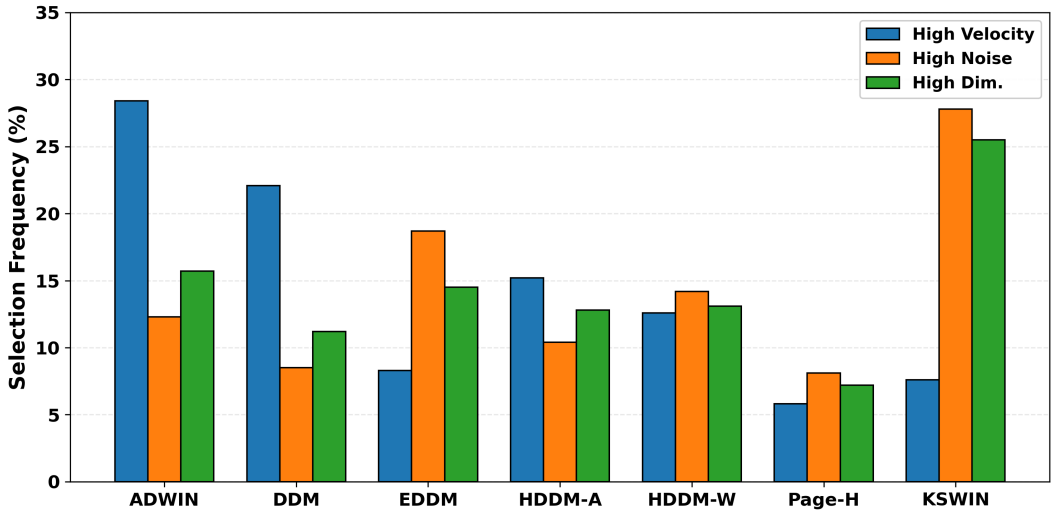


Figure 5: Detector selection frequency by stream characteristics. Selection patterns align with domain knowledge: ADWIN/DDM for high velocity, KSWIN for high noise/dimensionality.